

HUNTSVILLE INTL, AL, USA

WMO: 723230

Lat: 34.644N

Lon: 86.786W

Elev: 624

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03856

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.5	(c) 22.7	(d) 6.8	(e) 8.0	(f) 23.8	(g) 10.9	(h) 9.8	(i) 26.2	(j) 24.0	(k) 51.8	(l) 20.9	(m) 48.5	(n) 7.3	(o) 10	(p) 0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	95.4	74.9	93.2	74.6	91.1	74.1	78.6	88.2	77.7	87.3	76.8	86.0	7.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	140.2	82.6	75.2	135.1	81.7	74.1	130.4	80.9	42.5	88.3	41.4	87.2	40.5	86.1	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.1	(b) 17.8	(c) 16.0	(e) 11.9	(f) 98.2	(g) 5.0	(h) 3.2	(i) 8.3	(j) 100.6	(k) 5.4	(l) 102.4	(m) 2.6	(n) 104.3	(o) -1.1	(p) 106.6		
(a) 20.1	(b) 17.8	(c) 16.0	(e) 10.4	(f) 80.5	(g) 4.9	(h) 1.2	(i) 6.9	(j) 81.3	(k) 4.0	(l) 82.0	(m) 1.3	(n) 82.6	(o) -2.2	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	41.6	45.8	53.4	62.3	70.6	77.8	80.0	79.5	74.1	63.2	52.0	44.9
	DBStd	15.80	11.00	10.17	9.58	7.83	6.57	4.61	3.58	4.06	6.22	8.17	8.80	9.77
	HDD50	880	307	183	77	7	0	0	0	0	0	6	80	220
	HDD65	2969	726	539	371	140	28	0	0	0	10	133	399	623
	CDD50	5335	47	66	184	377	639	834	931	915	724	415	139	64
	CDD65	1949	1	2	13	60	202	384	466	450	284	77	8	2
	CDH74	18219	0	8	104	532	1800	3722	4503	4346	2518	638	45	3
	CDH80	7623	0	0	12	109	631	1657	2013	1995	1042	159	5	0
Wind	WSAvg	6.7	8.0	8.2	8.5	8.1	6.9	5.7	4.8	4.6	5.5	6.1	6.8	7.5
Precipitation	PrecAvg	56.1	5.1	4.8	6.5	4.6	5.2	4.3	4.6	3.4	4.0	3.1	4.7	5.7
	PrecMax	73.6	10.9	9.6	17.0	12.6	11.9	15.0	14.8	9.8	9.8	12.0	11.5	18.7
	PrecMin	41.8	1.3	0.6	1.6	0.4	1.3	0.2	0.8	0.6	0.5	0.0	0.6	0.7
	PrecStd	8.9	2.4	2.4	3.3	2.8	2.5	2.7	2.6	2.1	2.4	2.3	2.2	3.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	74.8	81.4	86.5	91.9	96.8	97.3	99.4	95.9	88.4	79.4	72.7
		MCWB	62.4	62.9	64.0	66.8	72.0	74.4	75.5	75.1	71.8	70.1	65.8	64.0
	2%	DB	66.3	70.1	76.8	83.2	89.0	93.9	94.4	95.3	92.5	84.1	73.9	68.3
		MCWB	60.4	61.4	62.2	65.7	70.7	74.0	76.2	75.5	72.4	68.6	62.2	61.7
	5%	DB	62.9	66.2	73.0	80.2	86.5	91.4	92.1	92.5	89.5	80.8	70.4	64.5
		MCWB	58.6	58.1	60.0	64.7	69.8	73.3	75.9	74.8	72.0	66.4	60.7	59.3
	10%	DB	59.2	62.2	69.2	76.7	83.7	89.0	89.8	89.8	86.2	77.4	67.0	60.6
		MCWB	54.2	55.2	59.1	63.1	68.7	73.0	75.4	74.5	71.1	65.1	59.2	55.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.7	66.7	67.8	71.3	75.2	78.7	80.3	79.8	76.9	74.8	69.2	66.3
		MCDB	68.6	71.3	74.3	80.9	84.0	88.3	89.8	90.3	86.9	81.0	73.8	70.7
	2%	WB	62.0	63.9	65.2	69.0	73.5	77.4	79.1	78.3	75.5	72.2	66.3	63.2
		MCDB	65.0	68.5	73.3	78.0	83.4	87.1	88.3	88.5	85.3	80.1	71.0	67.0
	5%	WB	59.3	60.3	62.8	67.3	72.3	76.2	78.1	77.3	74.5	69.9	63.6	60.1
		MCDB	62.8	64.3	70.2	75.8	82.2	85.9	87.4	87.2	83.7	77.0	68.6	63.7
	10%	WB	55.4	56.6	60.4	65.4	71.1	75.1	77.1	76.3	73.2	67.8	60.3	56.4
		MCDB	58.3	60.9	67.6	73.4	80.5	84.6	86.1	85.6	81.7	73.8	65.2	60.1
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.8	20.7	21.8	20.5	19.3	18.3	19.6	21.1	22.2	20.9	17.5
		MCDBR	19.7	23.2	25.0	24.9	23.4	22.7	21.4	23.2	24.2	25.1	24.6	20.0
		MCWBR	15.0	15.6	14.0	11.0	8.5	7.0	6.9	7.0	7.9	11.1	14.3	15.0
	5% WB	MCDBR	16.8	18.8	20.1	19.8	19.2	18.9	18.5	19.2	18.8	19.0	17.7	17.7
MCWBR		14.4	15.1	13.2	10.3	8.3	7.5	7.4	7.4	7.4	10.1	12.6	15.1	
Clear-Sky Solar Irradiance	taub	0.322	0.326	0.370	0.400	0.450	0.484	0.524	0.532	0.432	0.382	0.342	0.320	
	taud	2.479	2.465	2.367	2.276	2.181	2.120	2.029	1.993	2.261	2.383	2.439	2.503	
	Ebn at Noon	280	292	286	281	267	257	246	241	262	268	270	274	
	Edh at Noon	27	31	37	42	47	50	54	56	40	33	28	25	
All-Sky Solar Radiation	RadAvg	776	942	1310	1701	1866	1979	1888	1751	1517	1223	891	665	
	RadStd	60	117	123	117	145	155	104	94	133	158	95	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.31	N/A	N/A	+1.28	N/A	N/A	-112	-216	+367	+284
Regional (32 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+189

Nomenclature: See separate page